David Mohaisen, Professor
University of Central Florida
Department of Computer Science
Cybersecurity and Privacy Cluster

https://cs.ucf.edu/~mohaisen/

Grammar and Parsing for Natural Language

① Phrase Structure
② Context-Free Grammars
③ Structural Ambiguity
④ Dependency Grammar
⑤ Treebanks and Data
⑥ Neural Parsing Models

Introduction

Lecture Objectives

- Become familiar with dependency parsing
- Understand structure grammar
- Be familiar with context-free grammars
  - How CFG works in structure grammar
- Understanding sentence structure for accurate language interpretation and associated issues
- Dependency parsing as a tree structure example
- Deep learning-based parsing
- Related tools and examples.

Dependency Parsing

Dependency Parsing

- **Phrase Structure in Linguistics**
  - Phrase structure refers to the way words are organized into nested constituents that form larger, meaningful units within a sentence.
  - This hierarchical organization is fundamental to understanding the grammar and syntax of a language.

- **Starting Point: Words**
  - Every phrase structure begins with the smallest meaningful units in language- **words**. These serve as the building blocks for more complex structures.
  - **Individual words**: *The, cat, cuddly, by, door*

Dependency Parsing, cont.

- **Words Combine into Phrases**
  - When words come together, they form **phrases**, which are intermediary structures that convey more specific meaning.
  - These phrases are often categorized based on the role of the head word, such as noun phrases or prepositional phrases.
  - **Examples**:
    - *The cuddly cat* (a noun phrase)
    - *By the door* (a prepositional phrase)
- **Phrases Combine into Larger Phrases**
  - Phrases can combine or nest to create **larger, more complex phrases**, which increase the level of detail or clarify relationships between ideas in a sentence.
  - **Example**: *The cuddly cat by the door*
    - Here, *The cuddly cat* is a noun phrase, and *by the door* is a prepositional phrase that modifies it, providing additional information about the location of the cat.

Dependency Parsing, cont.

- **Why Phrase Structure Matters:** Understanding phrase structure is critical for:
  - **Sentence Parsing**: Breaking down sentences into their components for linguistic analysis.
  - **Grammar Rules**: Identifying how words and phrases must be arranged to create grammatical sentences.
  - **Language Learning**: Helping learners understand how words combine logically in a new language.

- This hierarchical organization underpins syntax and enables the creation of coherent sentences from simple elements.

Structure Grammar

Structure Grammar

- **Phrase structure grammar**, often equated with **context-free grammars (CFGs)**, is a formal system used in linguistics and computer science to describe the syntax of natural and formal languages.
- Grammar specifies how smaller elements (words or categories) combine to form larger structures (phrases or sentences) through a set of production rules.

- **Key Concepts of Phrase Structure Grammar**
  - **Context-Free Grammars (CFGs)**:
    1. CFGs operate on the principle that a grammatical rule applies independently of the surrounding context.
    2. These grammars define how syntactic categories (e.g., noun phrases, verb phrases) are structured and combined to form sentences.

Structure Grammar, cont.

Categories (Parts of Speech): Each word is assigned a **category** based on its syntactic role. These categories are foundational building blocks for creating phrases and sentences. The main categories are:

- **NOUN (N)**: Represents a person, place, thing, or idea. Ex.: *cat, door, freedom*
- **PRONOUN (PRON)**: Substitutes for a noun. Ex.: *he, she, it, they*
- **VERB (V)**: Denotes an action, occurrence, or state of being. Ex.: *run, is, jump*
- **ADJECTIVE (ADJ)**: Describes or modifies a noun. Ex.: *cuddly, large, blue*
- **ADVERB (ADV)**: Modifies a verb, adjective, or adverb. Ex.: *quickly, very, silently*
- **PREPOSITION (PREP)**: Links nouns or pronouns to other words, often showing relationships like direction or location. Ex.: *by, with, under*
- **CONJUNCTION (CONJ)**: Connects words, phrases, or clauses. Ex.: *and, but, or*
- **INTERJECTION (INT)**: Expresses emotion/exclamation. Ex.: *wow, ouch, hey*

How CFGs Work in Structure Grammar

- **Rules of Composition**: CFGs define rules that specify how categories combine to form phrases or sentences. For example:
  - **S → NP VP**: A sentence (S) consists of a noun phrase (NP) -> a verb phrase (VP).
  - **NP → DET N**: A noun phrase (NP) consists of a determiner (DET) -> a noun (N).
  - **VP → V NP**: A verb phrase (VP) consists of a verb (V) followed by a noun phrase (NP).
- **Recursive Nature**:
  - CFGs are recursive, allowing for the generation of nested structures. For instance:
    - *The cat by the door that is open*:
      The cat (NP) + *by the door that is open* (a prepositional phrase with embedded relative clause).
- **Example Analysis**:
  - Sentence: *The cuddly cat sat by the door.*
    - **S → NP VP**
    - **NP → DET ADJ N**: *The cuddly cat*
    - **VP → V PP**: *sat by the door*
    - **PP → PREP NP**: *by the door*

Understanding Structure for Interpretation

- Sentence structure is the foundation for interpreting language to convey meanings.
  - Without understanding the structure, language can become ambiguous, leading to misinterpretation or confusion: *"The boy saw the man with the telescope"* — Who has the telescope?
- **Communication of Complex Ideas:** Humans convey thoughts beyond simple statements by **combining words into larger units**. These combinations allow for:
  - **Precision**: Adding details (e.g., *"The tall building with glass windows"*).
  - **Relationships**: Clarifying interactions (e.g., subject-verb-object relationships).
- Larger structures like paragraphs and dialogues further enable the expression of nuanced and multi-layered ideas.
- **The Need to Understand Connections:** To interpret meaning, we identify:
  - **What is connected to what**: Understanding grammatical dependencies (e.g., *"The cat sat on the mat"* links the subject, action, and location).
  - **Hierarchical Relationships**: Determining the main clause versus modifiers or subclauses.
  - Recognizing these connections helps decode the logical flow and emphasis of a sentence, improving comprehension.

Propositional Phrase / Ambiguity

San Jose cops kill man with knife

Close

Text Paper Translate Listen

San Jose cops kill man with knife

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NEWS

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Science & Environment

Scientists count whales from space

By Jonathan Amos
BBC Science Correspondent

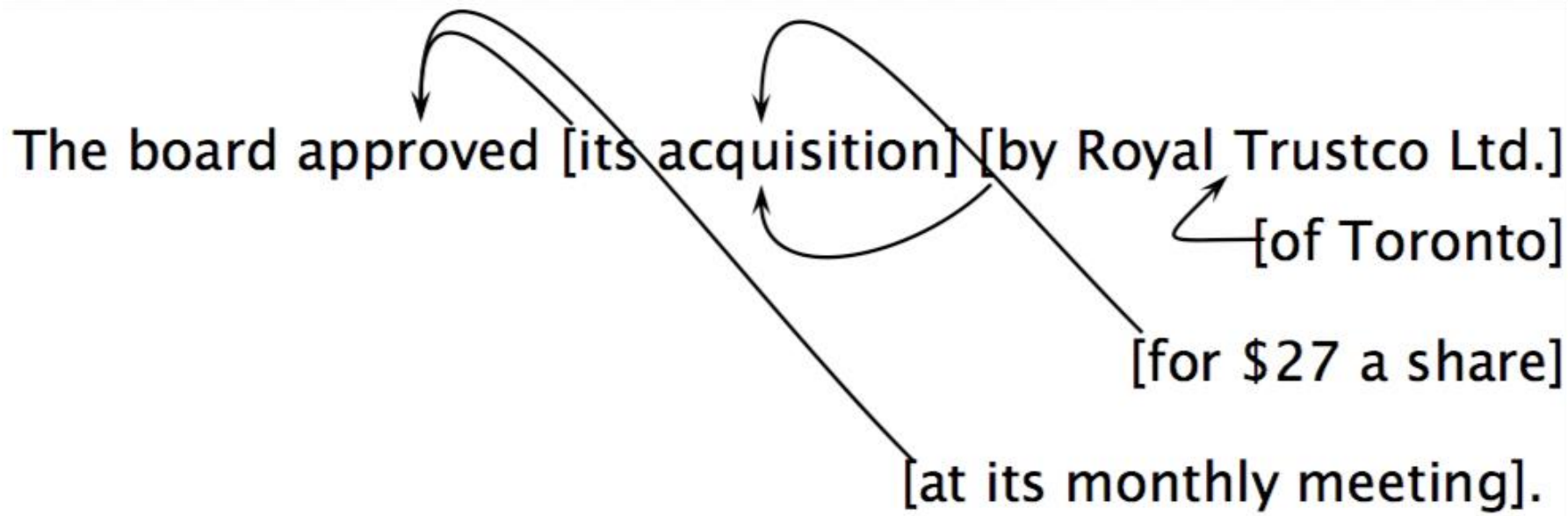
Scientists count whales from space

✓

Scientists count whales from space

PP Attachment Ambiguities

- A key parsing decision is how we 'attach' various constituents
  - PPs, adverbial or participial phrases, infinitives, coordination,



Coordination Scope Ambiguity

- **Shuttle veteran and longtime NASA executive Fred Gregory appointed to board**
- The sentence "**Shuttle veteran and longtime NASA executive Fred Gregory appointed to board**" can have different interpretations based on how the elements are connected. Here are some possible meanings:
  - **Basic Meaning: shuttle veteran** and a **longtime NASA executive**, has been appointed to a board.
  - **Dual Appointees:** Two individuals are implied: A **shuttle veteran and** a **longtime NASA executive** (Fred Gregory).
  - **Appointed to Two Boards:** Fred Gregory, described as both a **shuttle veteran** and a **longtime NASA executive**, is being appointed to **two boards**: One board for shuttle-related roles and Another board for executive leadership.
  - **Emphasis on Qualification** Fred Gregory's qualifications as both a **shuttle veteran** and a **NASA executive** are highlighted to justify his **appointment to a single board**.
  - **Organizational Appointment** Fred Gregory, as a **shuttle veteran** and **longtime NASA executive**, is not joining a corporate board but rather a NASA-related internal board.

Modifier Ambiguity

THE NEWS-GAZETTE

NATION / WO

PRESIDENT'S FIRST PHYSICAL

Doctor: No heart, cognitive issues

But Trump needs to reduce his cholesterol, lose weight

By JILL COLVIN

6-foot-3 president weighed in at 239 pounds — three pounds heavier than he was in September 2016, the last time Trump revealed his weight to the public.

Trump's blood pressure was 122 over 74, and his

with no medical issues." Trump has no heart disease and no family history of it.

The 71-year-old president performed "exceedingly well" on cognitive screening, which is not

White H
reporter

numbers, including some that featured a bucket and bells brigade of performers who beat rhythms on buckets and trash cans with drums sticks and hammer mallets. PHOTO BY JENNIFER STULTZ

MENTORING DAY

Students get first hand job experience

By Gale Rose
grose@pratttribune.com

Eager students invaded businesses all over Pratt Tuesday, October 24 as they looked for future job opportunities on Disability Mentoring Day.

The 97 students from 12 schools fanned out across Pratt and got first hand experience what it would be like to work at those 40 businesses. They asked questions and got some hands on experience with various operations.

Paola Luna of Pratt High School, Gina Patton of Kingman High School and America Fernandez of St. John chose the Main Street Small Animal Veterinarian Clinic for their business. Students got a tour of the facility, learned what happens in an examination, got to handle various animals and watched a snake eat a mouse.

Luna said she was interested in animal health and wanted to know more about caring for hurt animals. Patton likes all kinds of animals and said she learned a lot from the experience. Watching the snake eat the mouse impressed her the most.

Fernandez wants to become a veterinarian and enjoyed learning everything that veterinarians

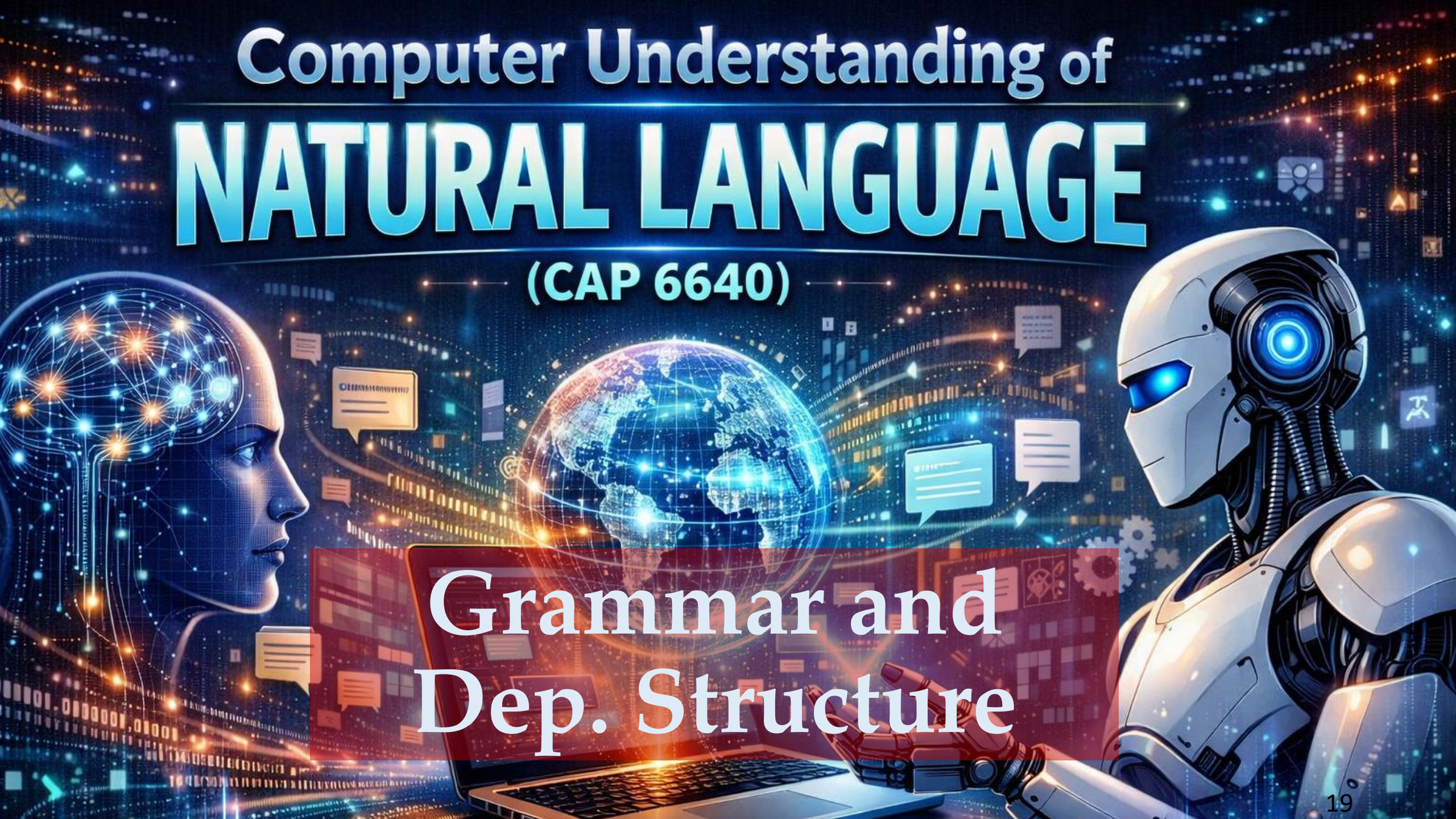
SEE MENTORING, 6

ug Meyer
ty Commissioner

- Hospital Pharmacist for 41 years
- 4 years Commissioner for Pratt Planning and Zoning Board of Appeals
- 3 years Pratt City Commission
- Graduate of Pratt High School and KU School of Pharmacy
- Past Member and President of Civic Groups and Organizations
- Experience and Knowledge of Financial Responsibility and Budgeting
- Supports Family Values, Education, and Business Growth
- Common Sense Approach for the Sustained Progress of Pratt

12 SATURDAY, October 28, 2017 ■ The Pratt Tribune ■ www.pratttribune.com

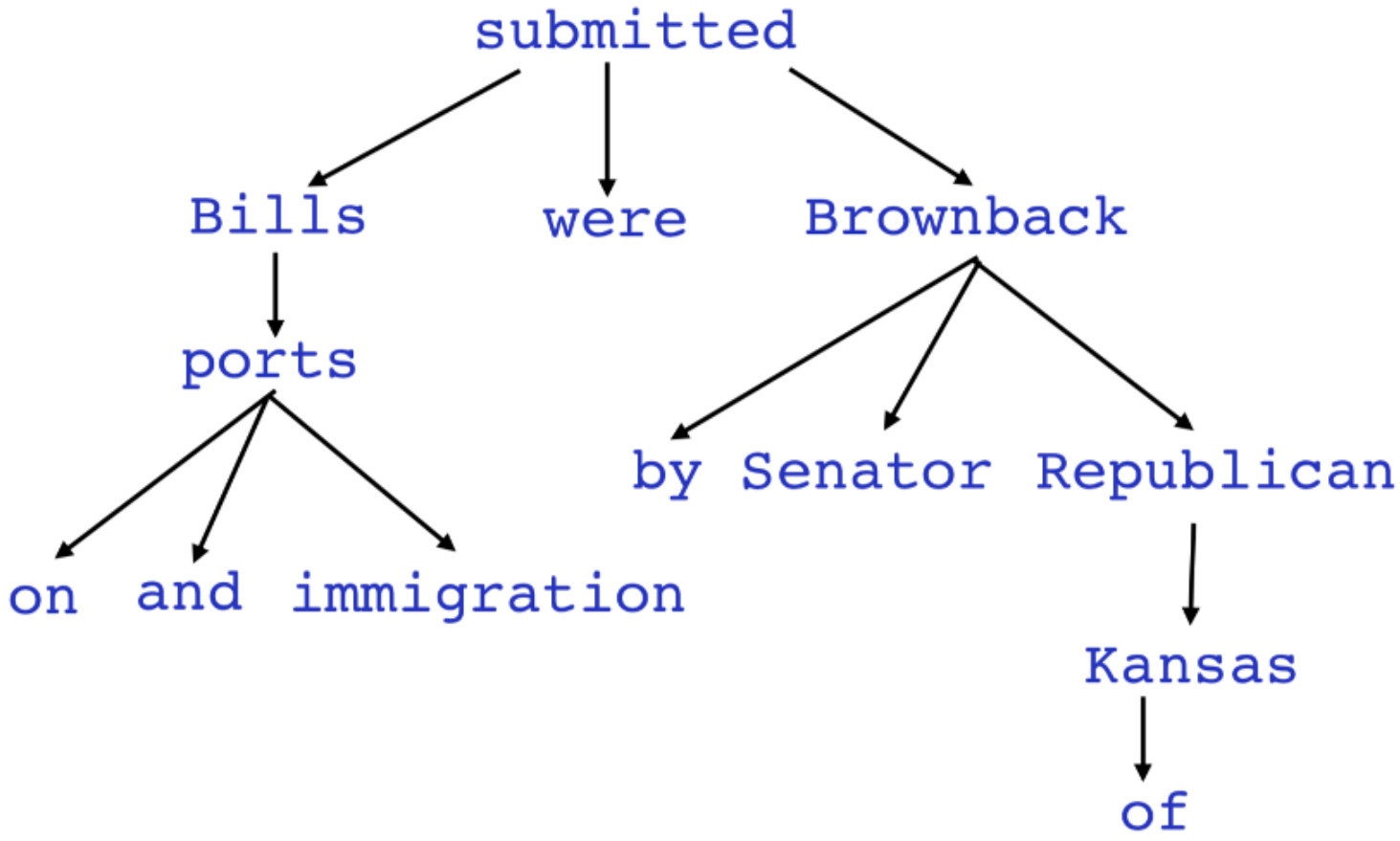
Modifier Ambiguity, cont.



Grammar and Dep. Structure

Dependency Grammar and Dep. Structure

- Dependency syntax postulates that syntactic structure consists of relations between lexical items, normally binary asymmetric relations (arrows) called dependencies



Core dependents of clausal predicates

Nominal dep	*Predicate dep*	
nsubj	csubj	
nsubjpass	csubjpass	
obj	ccomp	xcomp
iobj		

Non-core dependents of clausal predicates

Nominal dep	*Predicate dep*	*Modifier word*
	advcl	advmod
nmod		neg
↳ nmod:npmod		
↳ nmod:tmod		
↳ nmod:poss		

Special clausal dependents

Nominal dep	*Auxiliary*	*Other*
vocative	aux	mark
discourse	auxpass	punct
expl	cop	

Noun dependents

Nominal dep	*Predicate dep*	*Modifier word*
nummod	acl	amod
	↳ acl:relcl	
appos		det
		↳ det:predet
nmod		neg

Compounding and unanalyzed

compound	fixed	goeswith
↳ compound:prt		
flat	foreign	

Coordination

conj	cc	punct
	↳ cc:preconj	

Case-marking, prepositions, possessive

case

Loose joining relations

list	parataxis	orphan
dislocated		reparandum

Other

Sentence head	*Unspecified dependency*
root	dep

Relationships

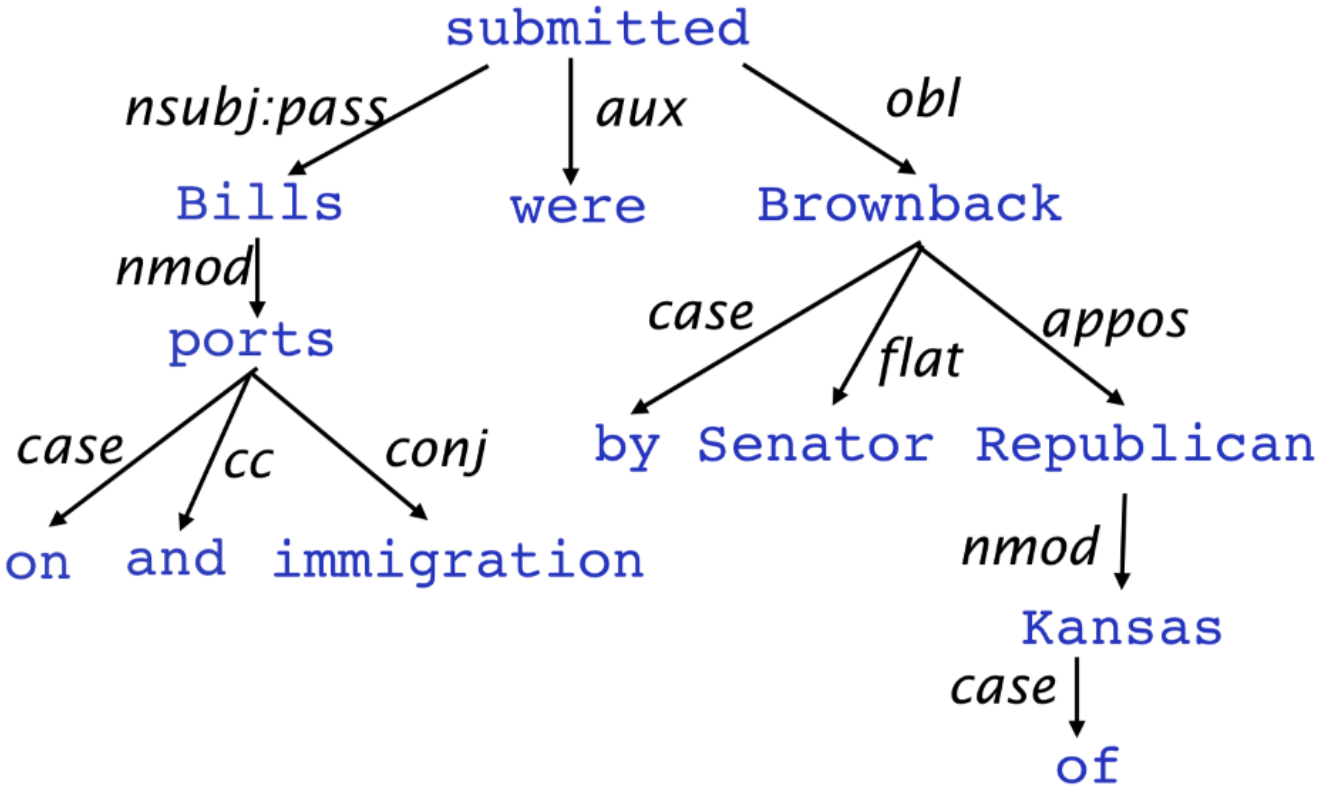
- **nsubj** – nominal subject (who is doing the action)
- **nsubjpass** – subject in passive voice
- **cop** – copula (linking verb like *is, was*)
- **det** – determiner (a, the, this)
- **amod** – adjectival modifier (adjective describing a noun)
- **compound** – compound noun (e.g., *goat trainer*)
- **ccomp** – clausal complement (embedded clause)
- **xcomp** – open clausal complement (no subject of its own)
- **advmod** – adverbial modifier
- **aux** / **auxpass** – auxiliary verb (is, was, do, etc.)
- **conj** – conjunction relation (and/or)
- **cc** – coordinating conjunction (*and, or*)
- **nmod** – nominal modifier (often prepositional phrases)
- **case** – preposition or case marker (*in, on, by*)
- **expl** – expletive (dummy words like *there* in "there are")
- **punct** – punctuation

Dependency Grammar and Dep. Structure, cont.

- Dependency syntax postulates that syntactic structure consists of relations between lexical items, normally binary asymmetric relations (arrows) called dependencies

The arrow connects a head (governor, superior, regent) with a dependent (modifier, inferior, subordinate)

Usually, dependencies form a tree (connected, acyclic, single-head)



Dependency Grammar History

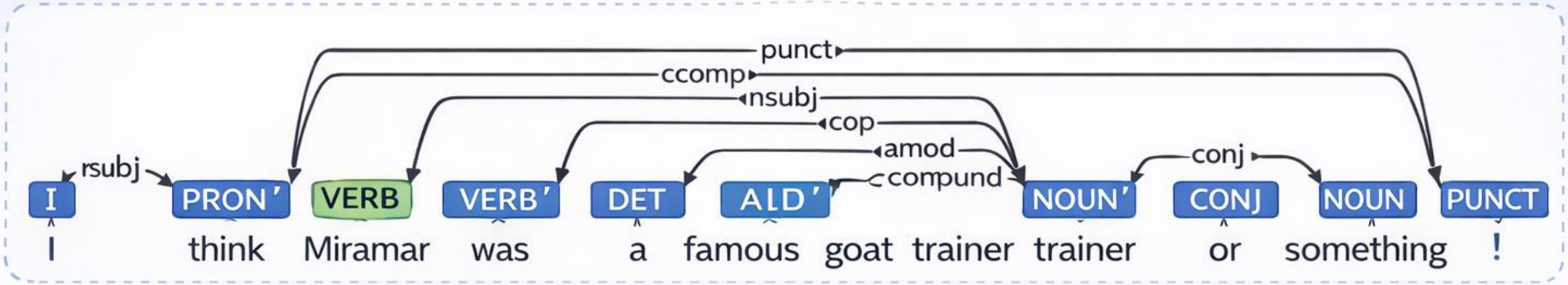
- **Origins**: Dependency-based analysis dates back to ancient grammatical traditions:
  - **Sanskrit Grammar (Pāṇini)**: First systematic linguistic framework (~4th Century BCE).
  - **Arabic Grammar (Sibawayh)**: Explored hierarchical word relationships (~8th Century CE).
- **Early European Development**:
  - **Lucien Tesnière (1959)**: Coined *Dependency Grammar* in his work *"Éléments de syntaxe structurale"*. Introduced concepts like the *governor-dependent* relationship and syntactic trees.
- **Modern Computational Linguistics**:
  - Dep. grammar gained prominence for simplicity and suitability for computational parsing.
  - Influenced modern **dep. parsers** used in NLP, such as Stanford and Universal Dependency.

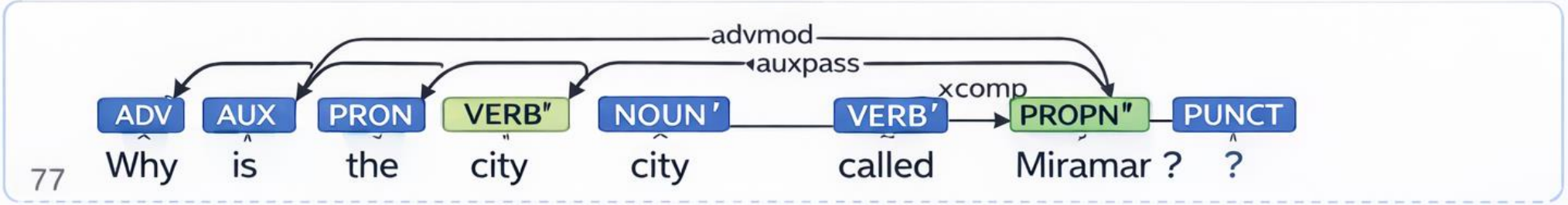
- **Key Advantages**:
  - Emphasizes *word-to-word relationships* over phrase-level groupings (in constituency grammar).
  - Facilitates efficient representation of non-projective and free-word-order languages.

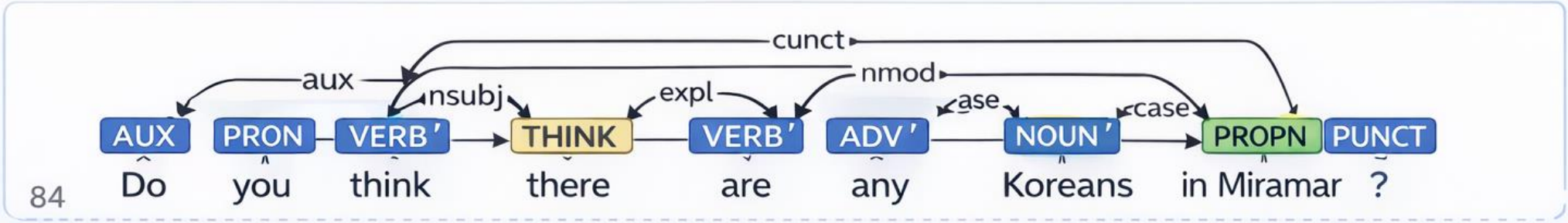
Dependency Grammar and Dep. Structure

- Some people draw the arrows one way; some the other way!
  - Tesnière had them point from head to dependent…
- Usually add a fake **ROOT** so every word is a dependent of precisely 1 other node

Dependency Parsing Example







PyTorch Example

```python
import torch
# Define the grammar rules
grammar_rules = {
    "S": [["NP", "VP"]],      # Sentence (S) = Noun Phrase (NP) + Verb Phrase (VP)
    "NP": [["DET", "N"], ["PRON"]], # Noun Phrase (NP) = Determiner (DET) + Noun (N) or
Pronoun (PRON)
    "VP": [["V", "NP"], ["V"]],  # Verb Phrase (VP) = Verb (V) + NP or just Verb (V)
    "DET": [["the"], ["a"]],
    "N": [["cat"], ["dog"], ["door"]],
    "PRON": [["he"], ["she"], ["they"]],
    "V": [["saw"], ["sat"], ["jumped"]],
}
# Vocabulary and word-to-index mapping
vocab = {
    "the": 0, "a": 1, "cat": 2, "dog": 3, "door": 4,
    "he": 5, "she": 6, "they": 7, "saw": 8, "sat": 9, "jumped": 10
}
index_to_word = {idx: word for word, idx in vocab.items()}

# Example sentence to parse
sentence = ["the", "cat", "sat"]
# Convert sentence to tensor
sentence_tensor = torch.tensor([vocab[word] for word in sentence])
# Define a parsing function
def parse(cfg, sentence):
    def match_rule(rule, words):
        if not rule or not words:
            return len(rule) == len(words) == 0
        if rule[0] in cfg:  # Non-terminal
            for expansion in cfg[rule[0]]:
                if match_rule(expansion + rule[1:], words):
                    return True
        elif rule[0] == words[0]:  # Terminal
            return match_rule(rule[1:], words[1:])
        return False

    for start_rule in cfg["S"]:
        if match_rule(start_rule, sentence):
            return True
    return False
# Parse the sentence
is_valid = parse(grammar_rules, sentence)
print(f"Is the sentence '{' '.join(sentence)}' valid? {'Yes' if is_valid else 'No'}")
```

- **Grammar Definition**:
  - The grammar_rules dictionary specifies the context-free grammar, with each non-terminal symbol (S, NP, etc.) associated with a list of its possible expansions.
- **Vocabulary**:
  - The vocab dictionary maps each word to an index, which is used to convert the sentence into a tensor for processing.
- **Sentence Parsing**:
  - The parse function recursively checks if the sentence matches the CFG by trying all possible expansions of the rules.
- **Example**:
  - For input sentence ["the", "cat", "sat"], the function evaluates whether it can be generated by the grammar, returning True if valid.

Annotated Data

Annotated Data

- Starting off, building a treebank seems a lot slower and less useful than building a grammar
- However, treebanks give us many things
  - Reusability of the labor
    - Many parsers, part-of-speech taggers, etc. can be built on it.
    - Valuable resource for linguistics.
  - Broad coverage, not just a few intuitions.
  - Frequencies and distributional information.
  - A way to evaluate other NLP systems.

Deep Learning-based Parsing

- Deep learning-based grammar parsing, or **neural parsing**, uses deep neural nets to predict the syntactic structure of a sentence.
- Deep learning learn the structure of language from large datasets, enabling them to handle complex and ambiguous sentences more effectively.
  - **Data-Driven Learning**: Instead of relying on handcrafted rules, deep learning models learn directly from labeled datasets of sentences annotated with their syntactic structures (e.g., dependency trees or constituency trees).
  - **Representation Learning**: Words and phrases are represented as vectors (embeddings) in a continuous space. Contextual word embeddings (e.g., from models like BERT or GPT) capture the meaning of words in context.
- **Parsing Approaches**: Dependency and constituency parsing
  - **Dependency Parsing**: Predicts grammatical relationships b/w words in a sentence.
  - **Constituency Parsing**: Divides a sentence into nested subphrases, forming a tree.
- **Neural Architectures**: Modern parsers encode input sentences and predict syntactic structures using RNNs (for sequential dependency), Transformers (for long-range dependency), and GNNs (for tree-like structures).

Steps of Deep Learning-based Parsing

1. **Input Representation**: Sentences are tokenized and converted into embeddings. Pre-trained models like BERT or RoBERTa are often used to generate contextualized word embeddings.
2. **Encoding Sentence Context**: The sequence of word embeddings is passed through an encoder (e.g., Transformer) to capture the relationships between words in context.
3. **Score Prediction**:
    1. For dependency parsing: The model predicts scores for possible edges between words to form a dependency tree.
    2. For constituency parsing: The model predicts scores for splitting points in the sentence to recursively build a hierarchical tree.
4. **Tree Construction**: Using the predicted scores, the parser constructs the syntactic tree. Techniques like the **Eisner algorithm** (for dependency parsing) or chart parsing (for constituency parsing) may be used to ensure valid tree structures.
5. **Training**: Models are trained on large datasets annotated with syntactic trees, optimizing a loss function (e.g., cross-entropy loss) that measures the difference between predicted and true structures.

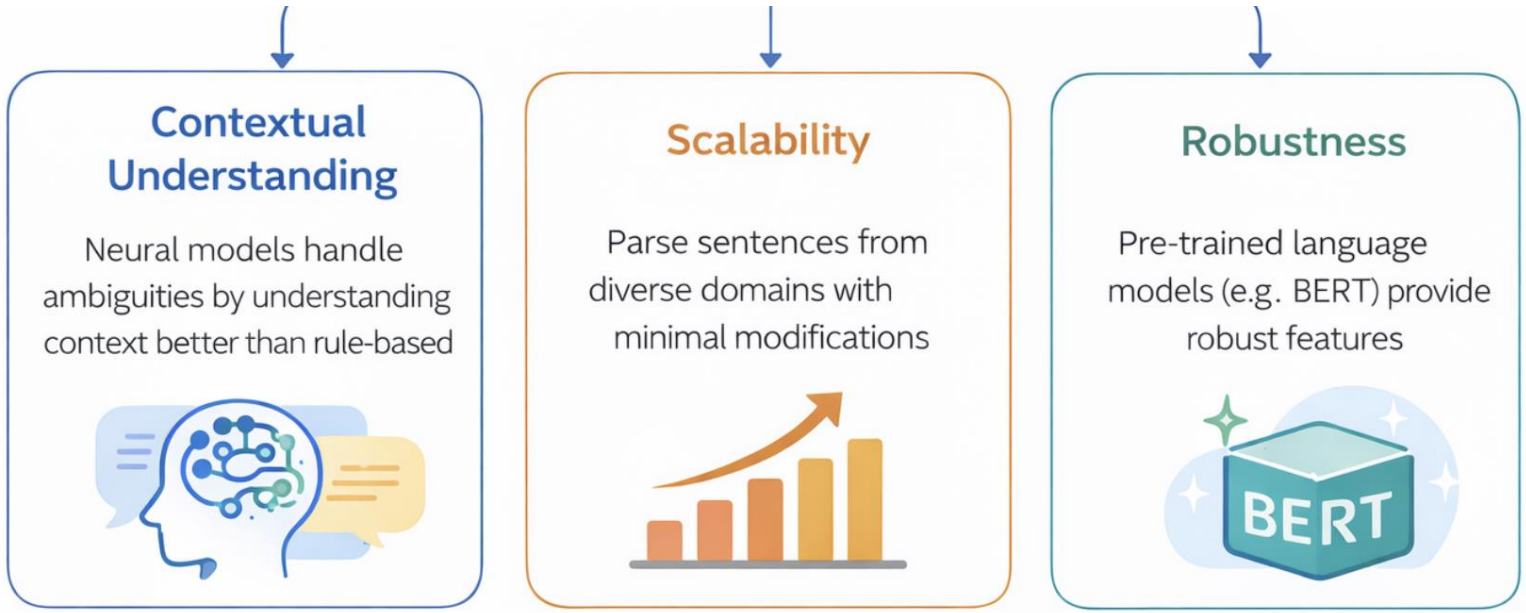
Modern Parsing

Modern Parsing Models

- **BiLSTM + MLP Parsers**:
  - Combine bidirectional LSTMs for sequence encoding with multi-layer perceptrons (MLPs) for edge prediction in dependency parsing.
  - Example: Stanford Neural Dependency Parser.
- **Transformer-Based Models**:
  - Leverage pre-trained transformer encoders (e.g., BERT) for feature extraction, followed by specialized layers for parsing.
  - Example: Benepar (Berkeley Neural Parser), which uses BERT for constituency parsing.
- **Graph-Based Parsers**:
  - Treat sentences as graphs and use graph neural networks (GNNs) to predict dependencies.
  - Example: Models leveraging GCNs for parsing tasks.

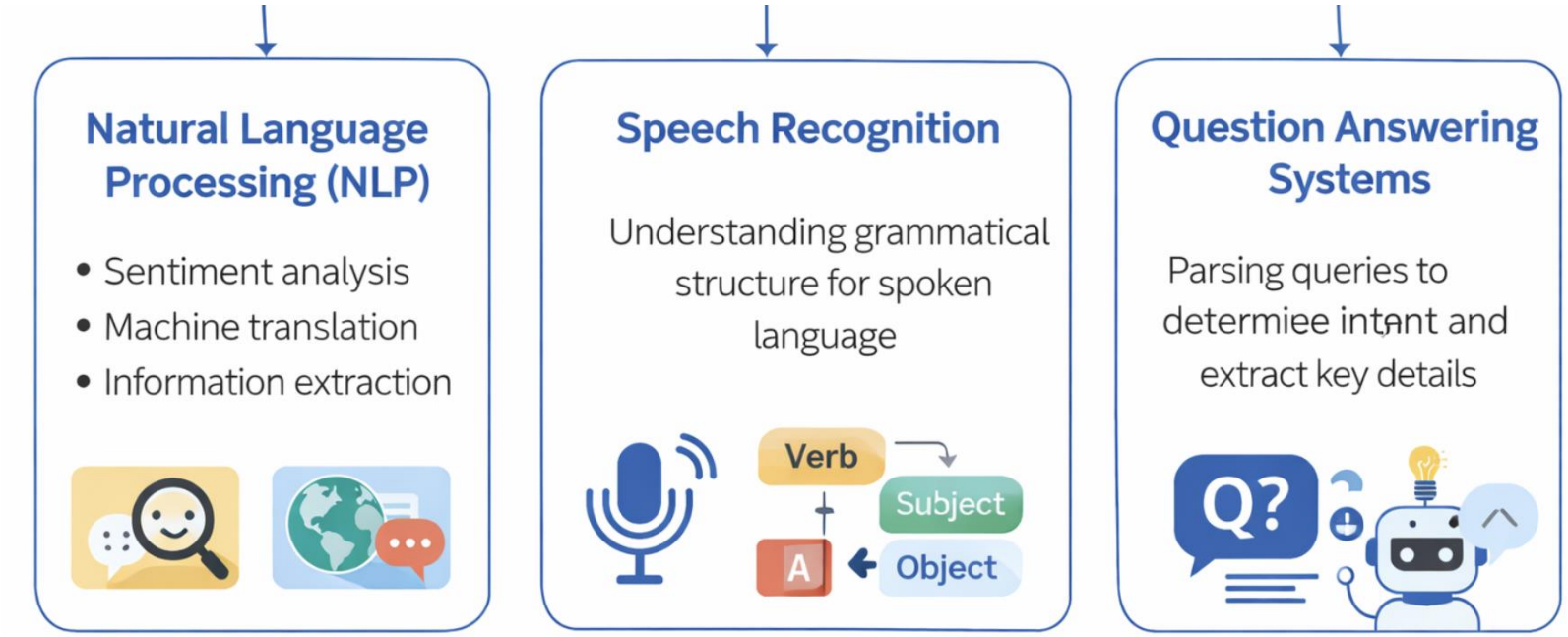
Advantages of Deep Learning-Based Parsing

- **Contextual Understanding**: Neural models handle ambiguities by understanding the context of words and phrases better than rule-based.
- **Scalability**: These models can parse sentences from diverse domains (e.g., news, social media) with minimal modifications.
- **Robustness**: Pre-trained language models (e.g., BERT) provide robust features, improving parsing performance significantly.



Applications

- **Natural Language Processing (NLP)**: Sentiment analysis, machine translation, and information extraction.
- **Speech Recognition**: Understanding structure for spoken language.
- **Question Answering Systems**: Parsing queries to determine intent and extract key details.



Example Libraries for Neural Parsing

1. **SpaCy**: Offers dependency parsing with pre-trained neural models.
2. **AllenNLP**: Supports advanced constituency and dependency parsing.
3. **Stanza**: Provides multilingual neural parsing (by Stanford NLP).
4. **Transformers (Hugging Face)**: Models like BERT or GPT can be adapted for parsing tasks.

- By leveraging the ability of deep learning to model complex patterns, neural parsing systems achieve state-of-the-art results in syntactic and semantic understanding tasks.

Q&A